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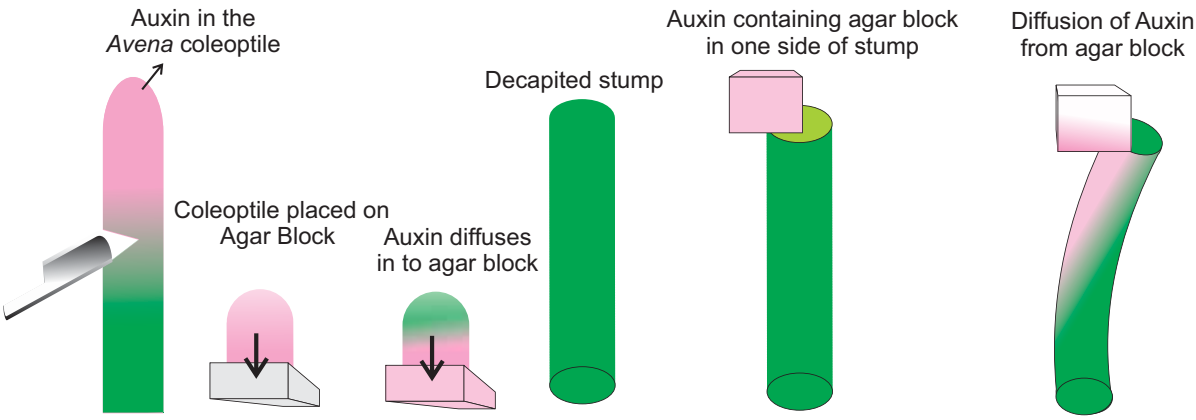


Figure 15.12: *Avena* Curvature Test

placed on one side of the decapitated stump of *Avena* coleoptile. The auxin from the agar blocks diffuses down through coleoptile along the side to which the auxin agar block is placed. An agar block without auxin is placed on another decapitated coleoptile. Within an hour, the coleoptiles with auxin agar block bends on the opposite side where the agar block is placed. This curvature can be measured (Figure 15.12).

10. Physiological Effects

- They promote cell elongation in stem and coleoptile.
- At higher concentrations auxins inhibit the elongation of roots but induce more lateral roots. Promotes growth of root only at extremely low concentrations.
- Suppression of growth in lateral bud by apical bud due to auxin produced by apical bud is termed as **apical dominance**.
- Auxin prevents abscission.
- It is responsible for initiation and promotion of cell division in cambium, which is responsible for the secondary growth and tumor. This property of induction of cell division has been exploited for tissue culture techniques and for the formation of callus.
- Auxin stimulates respiration.
- Auxin induces vascular differentiation.

Agent Orange

Mixture of two phenoxy herbicides 2,4-D and 2,4,5-T is given the name 'Agent orange' which was used by USA in Vietnam war for defoliation of forest (chemical warfare).



In botanical gardens and tea gardens, gardeners trim the plants regularly so that they remain bushy. Does this practice have any scientific explanation?

Yes, trimming of plants removes apical buds and hence apical dominance. The lateral buds sprout and make the plants bushy.